

Algebra MATH-310

Lecture 3

Anna Lachowska

September 28, 2026

Plan of the course

- 1 Integers: 1 lecture
- 2 Groups: 6 lectures
- 3 Rings and fields: 5 lectures
- 4 Review: 1 lecture

Today: Groups: lecture 2

- a Recall: basic examples of groups
- b Group homomorphisms and isomorphisms
- c Presentation of a group in generators and relations
- d Examples of group homomorphisms
- e Kernel and image of a group homomorphism

Recall: Groups: definition

Definition

A **group** is a set G with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the axioms:

- 1 Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for any $a, b, c \in G$.
- 2 Neutral element: $\exists 1 \in G$ such that $1 \cdot a = a \cdot 1 = a$ for any $a \in G$
- 3 Inverse: For any $a \in G \exists a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1}a = 1$.

Definition

A group G is called **finite** if $|G| < \infty$. In this case $|G| \in \mathbb{N}$ is called **the order of the group**.

Definition

A group G is called **abelian** if $a \cdot b = b \cdot a$ for any $a, b \in G$.

Recall: two groups modulo n

Additive group modulo n

For any $n \in \mathbb{N}$, $n \geq 2$, the equivalence classes of integers modulo n : $(\mathbb{Z}/n\mathbb{Z}, +, 0) = \{[0], [1], \dots, [n-1]\}$ form an abelian group with respect to addition. $|(\mathbb{Z}/n\mathbb{Z}, +, 0)| = n$.

Multiplicative group modulo n

For any $n \in \mathbb{N}$, $n \geq 2$, define the group $((\mathbb{Z}/n\mathbb{Z})^*, \cdot, 1) = \{x \in \mathbb{N} : 1 \leq x \leq n, \gcd(x, n) = 1\}$. Then $((\mathbb{Z}/n\mathbb{Z})^*, \cdot, 1)$ is an abelian group with respect to multiplication and $|(\mathbb{Z}/n\mathbb{Z})^*, \cdot, 1| = \varphi(n)$.

Recall: every element of $((\mathbb{Z}/n\mathbb{Z})^*, \cdot, 1)$ has a multiplicative inverse.

$$\gcd(a, n) = 1 \iff \exists x, y \in \mathbb{Z} : ax + ny = 1$$

$$[a][x] + \underbrace{[n][y]}_{[0]} = [1] \bmod n \iff [a] \cdot [x] = [1] \Rightarrow [a]^{-1} = [x]$$

Recall: Subgroups, cosets, Lagrange's and Euler's theorems

Definition

A **subgroup** $H \subset G$ is a subset in G such that $1 \in H$ and H is closed with respect to the multiplication and taking inverses.

Definition

Let $H \subset G$ be a subgroup and $g \in G$ an element. The **left coset** gH is the set of group elements of the form $gH = \{gh, h \in H\}$.

Theorem

(Lagrange's theorem)

Let G be a finite group and $H \subset G$ a subgroup. Then $|H|$ divides $|G|$.

Theorem *if p is a prime that doesn't divide $a \Rightarrow a^{p-1} \equiv 1 \pmod{p}$*
 $\gcd(a, p) = 1$

(Euler's theorem)

Let $a, n \in \mathbb{Z}_+$ such that $\gcd(a, n) = 1$. Then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Group of the roots of unity

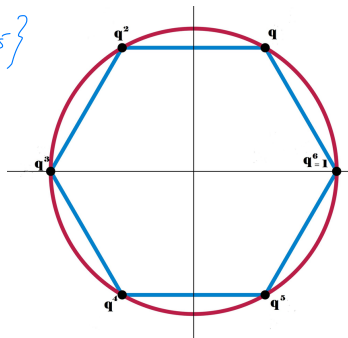
Let $n \in \mathbb{N}$, $n \geq 2$ and consider the group $C_n = \{1, q, q^2, \dots, q^{n-1}\}$ where $q = e^{\frac{2\pi i}{n}}$.

Ex. C_6

$$\{1, q = e^{\frac{2\pi i}{6}}, q^2, q^3, q^4, q^5\}$$

$$q^3 \cdot q^5 = q^2$$

$$|C_6| = 6$$



Inverses:

$$q \cdot q^5 = 1$$

$$q^2 \cdot q^4 = 1$$

$$q^3 \cdot q^3 = 1$$

Intuition: the groups $(\mathbb{Z}/n\mathbb{Z}, +, 0)$ and C_n are "the same"

For any $n \in \mathbb{N}, n \geq 2$,

$(\mathbb{Z}/n\mathbb{Z}, +, 0) = \{[0], [1], \dots, [n-1]\}$ is an abelian group of order n with respect to addition.

$C_n = \{1, q, q^2, \dots, q^{n-1}\}$ where $q = e^{\frac{2\pi i}{n}}$ is an abelian group of order n with respect to multiplication.

$$|\mathbb{Z}/n\mathbb{Z}, +, 0| = n \quad |C_n| = n$$

$$\begin{array}{ccc} \begin{array}{c} + \\ [0] \end{array} & \longrightarrow & \begin{array}{c} \cdot \\ 1 \end{array} \\ [1] & \longrightarrow & q = e^{\frac{2\pi i}{n}} \\ [k] & \longrightarrow & q^k \end{array} \quad \text{The map respects the group operation!}$$

How to tell if two groups are "the same"?

Example: $|G_1| = |G_2|$

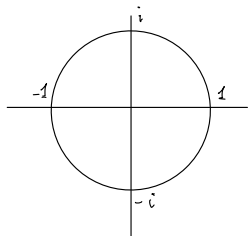
G_1 = $\{1, a, b, ab = ba \mid a^2 = b^2 = 1\}$. *Klein group*

$$aba = aab = a^2b = b$$

$$bab = bba = b^2a = a$$

$$\underbrace{abab}_{ab} = aabb = a^2b^2 = 1$$

$$|G_1| = 4$$



G_2 = $\{1, q, q^2, q^3 \mid q^4 = 1\}$. = $C_4 = \{1, q, q^2, q^3\}$

$$|G_2| = 4$$

$$\exists q = i \text{ s.t. } q^4 = 1 \text{ but } q^2 \neq 1$$

$$\text{But in } G_1, a^2 = 1, b^2 = 1, (ab)^2 = abab = 1$$

Groups G_1 and G_2 have different structure

Conclusion:

$|G_1| = |G_2|$ does not imply that the groups have the same structure

Group homomorphisms

Definition

A map $\phi : G \rightarrow H$ between two groups is a **group homomorphism** if

$$\phi(x \cdot_G y) = \phi(x) \cdot_H \phi(y) \quad \forall x, y \in G.$$

Proposition

If $\phi : G \rightarrow H$ is a group homomorphism, then $\phi(1_G) = 1_H$ and $\phi(x^{-1}) = (\phi(x))^{-1}$ for any $x \in G$.

Proof: Let $x, y \in G \Rightarrow \phi(\underbrace{x \cdot y^{-1}}_{1_G}) = \phi(x \cdot y^{-1}) \cdot \phi(y) = \phi(x) \quad \forall x, y \in G$
 $\Rightarrow \phi(x \cdot y^{-1}) = \phi(x) \cdot (\phi(y))^{-1}$

Take $y = x \Rightarrow \phi(\underbrace{x \cdot x^{-1}}_{1_G}) = \phi(x) \cdot (\phi(x))^{-1} = 1_H \Rightarrow \phi(1_G) = 1_H$
 $\phi(x) \cdot \phi(x^{-1}) = 1_H \Rightarrow \phi(x^{-1}) = (\phi(x))^{-1}$



Group homomorphisms

Definition

A group homomorphism that is invertible is called **an isomorphism**:

$$\phi : G \rightarrow H, \quad \psi : H \rightarrow G : \quad \phi \circ \psi = \text{Id}_H, \quad \psi \circ \phi = \text{Id}_G.$$

Then $G \simeq H$ are **isomorphic groups**.

A **group automorphism** is an isomorphism of a group to itself $\phi : G \rightarrow G$.

Example of a group isomorphism: $(\mathbb{Z}/n\mathbb{Z}, +, 0) \simeq C_n$

$$C_n = \{1, g, \dots, g^{n-1}\}$$

$$\varphi: \begin{array}{cccc} 1 & g & g^2 & g^{n-1} \end{array} \quad \text{is a bijection}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ [0] & [1] & [2] & [n-1] \end{array}$$

$$\varphi(g^i g^j) = \varphi(g^{i+j}) = [i+j]$$

$$\varphi(g^i) + \varphi(g^j) = [i] + [j] \pmod{n}$$

$$\varphi(1) = \varphi(\underbrace{g \cdot g \cdot \dots \cdot g}_n) = \underbrace{\varphi(g) + \dots + \varphi(g)}_n = \underbrace{[1] + [1] + \dots + [1]}_n = [n] = [0]$$

$$\varphi: C_n \rightarrow \mathbb{Z}/n\mathbb{Z} \quad \text{is a group isomorphism}$$

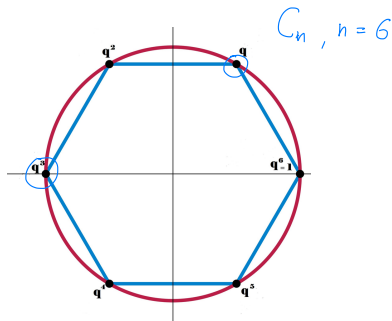
$$C_n \simeq \mathbb{Z}/n\mathbb{Z} \quad \text{isomorphic groups}$$

Conclusions

- 1 A **group homomorphism** is a map between two groups that respects the multiplication.
- 2 Two **isomorphic groups** may have different descriptions but they admit a bijective map (an isomorphism) that respects the group operation. In particular, it sends the product to product and inverse to inverse.

$\mathbb{Z}/6\mathbb{Z}, +$

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4



Presentation of a group in generators and relations

Definition

Generators of a group is a minimal set of elements of G such that any element of G can be written as a product of the generators and their inverses.
group operation

Example

$C_n = \{1, q, q^2, \dots, q^{n-1}\}$ has a generator q .

Can it have any other generators?

$$\gcd(k, n) = 1 \Leftrightarrow \exists a, b \in \mathbb{Z} : ak + bn = 1 \Leftrightarrow (q^k)^a = q \Rightarrow$$

it generates C_n

For example q^{n-1} is a generator

In general any q^k with $\gcd(k, n) = 1$ can be a generator.

Generators: example

Let $\mathbb{Q}_+^*$ denote the set of positive rational numbers. Then $\mathbb{Q}_+^*$ is a group with respect to multiplication.

Poll: What are the generators of the group $\mathbb{Q}_+^*$?

- A:** All squares of prime numbers and 1 *cannot get 2, only 2^{2k-2m}*
- B:** All positive natural numbers $\mathbb{N}_+$ *too many : $6 = 2 \cdot 3$*
- C:** All odd prime numbers and $\frac{1}{2}$ *$n = p_1^{m_1} \dots p_k^{m_k}$; $2 = (\frac{1}{2})^{-1}$; $\frac{1}{p_1^{m_1}} \dots \frac{1}{p_k^{m_k}}$ get all $\mathbb{Q}_+^*$*
- D:** All odd positive integers and 2 *too many: $9 = 3 \cdot 3$*
- E:** All numbers of the form $\frac{1}{n}$ where $n \in \mathbb{N}_+$ *too many: $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$*

Relations in a group

Definition

Any equation satisfied by the products of generators is called a **relation** in a group.

Example

In $C_n = \{1, q, q^2, \dots, q^{n-1}\}$ we have a generator q that satisfy the relation $q^n = 1$.
 $q^{n+3} = q^3$ is another relation.

Generators and relations in a group are not unique in general!

Group defined by generators and relations

Definition

A presentation of G in terms of generators and relations is an expression $\langle S \mid R \rangle$, where S is a set of generators and R is a minimal set of relations in G , such that any other relation in G follows from these.

Example C_n cyclic group of order n

$$C_n = \langle q \mid q^n = 1 \rangle.$$

Example: The Klein gp $K = \langle \underbrace{a, b}_S \mid \underbrace{a^2=1, b^2=1, ab=ba}_R \rangle$

Another possible presentation of the same gp $K = \langle \underbrace{a, b}_S \mid \underbrace{a^2=1, b^2=1, (ab)^2=1}_R \rangle$
if $ab=ba \Rightarrow (ab)^2 = abab = aabb = a^2b^2 = 1$
if $(ab)^2 = 1 \Rightarrow abab = 1 \Rightarrow bab = a^{-1} = a \Rightarrow bab^2 = ab \Rightarrow ba = ab$

What are they useful for?

Defining group homomorphism in terms of generators and relations

Proposition

Let $G = \langle S \mid R_1, R_2, \dots, R_k \rangle$ be a group defined by generators and relations, and H another group. Define $\phi : G \rightarrow H$ as follows

- (a) Define $\phi(s_i)$ for each generator $s_i \in S$ and $\phi(1_G) = 1_H$.
- (b) Set $\phi(x_1 \cdot x_2) = \phi(x_1) \cdot \phi(x_2)$, $\phi(x_1^{-1}) = (\phi(x_1))^{-1}$ for any $x_1, x_2 \in S$ *and their products*
- (c) Then $\phi : G \rightarrow H$ is a group homomorphism if and only if the defining relations $R_1, \dots, R_k$ in G are satisfied by the images of the generators $\phi(s_i)$ in H .

Idea: If R_2 is not satisfied by $\phi(s_i) \Rightarrow$

$$\phi(R_2) = \phi(s_1 \dots s_k) = \phi(1_G)$$

$$R_2: s_1 \dots s_k = 1 \quad \text{But } \phi(s_1) \dots \phi(s_k) \neq 1_H \Rightarrow \phi(1_G) \neq 1_H$$

ϕ is not a group homomorphism

Examples of group homomorphisms

Let $C_8 = \langle q \mid q^8 = 1 \rangle$, $C_4 = \langle t \mid t^4 = 1 \rangle$.

Let $\phi : C_8 \rightarrow C_4$ be a group homomorphism. Need to define: $\phi(q) \in C_4$

$\phi(q) = t^k$ Needs to satisfy: $(\phi(q))^8 = 1 \Rightarrow (t^k)^8 = 1 \Rightarrow t^{8k} = 1$
 $\phi(q^8) = \phi(1)$

$t^{8k} = 1$, true for all k since $t^{8k} = (t^4)^{2k} = 1^{2k} = 1$ in C_4
 $\Rightarrow k = 0, 1, 2, 3$ 4 homomorphisms

		1	q	q^2	q^3	q^4	q^5	q^6	q^7
ϕ_1	$k = 1$	1	t	t^2	t^3	1	t	t^2	t^3
ϕ_2	$k = 2$	1	t^2	1	t^2	1	t^2	1	t^2
ϕ_3	$k = 3$	1	t^3	t^2	t	1	t^3	t^2	t
ϕ_0	$k = 0$	1	1	1	1	1	1	1	1

Examples of group homomorphisms

Let $C_8 = \langle q \mid q^8 = 1 \rangle$, $C_4 = \langle t \mid t^4 = 1 \rangle$.

Let $f : C_4 \rightarrow C_8$ be a group homomorphism.

$f(t) = q^k$, the condition to satisfy: $(f(t))^4 = (q^k)^4 = q^{4k} = 1$
 $\Rightarrow k$ has to be even $\Rightarrow 4$ homomorphisms

		1	t	t^2	t^3
f_0	$k = 0$	1	1	1	1
f_2	$k = 2$	1	q^2	q^4	q^6
f_4	$k = 4$	1	q^4	1	q^4
f_6	$k = 6$	1	q^6	q^4	q^2

Kernel and image of a group homomorphism

Definition

The **kernel** of a group homomorphism $\phi : G \rightarrow H$ is the set of elements $g \in G$ such that $\phi(g) = 1 \in H$.

Example: $f_4 : C_4 \rightarrow C_8$ $\ker f_4 = \{1, t^2\} \subset C_4$
 $\varphi_1 : C_8 \rightarrow C_4$ $\ker \varphi_1 = \{1, g^4\} \subset C_8$

Definition

Let $\phi : G \rightarrow H$ be a group homomorphism. Then $\phi(G) \subset H$ is called the **image** of ϕ .

Example: $f_4 : C_4 \rightarrow C_8$ $\text{Im } f_4 = \{1, g^4\} \subset C_8$
 $\varphi_1 : C_8 \rightarrow C_4$ $\text{Im } \varphi_1 = \{1, t, t^2, t^3\} = C_4$

Properties of the kernel and the image

Proposition

Let $\phi : G \rightarrow H$ be a group homomorphism.

Then $\ker \phi \subset G$ is a subgroup in G , and $\text{Im} \phi \subset H$ is a subgroup in H .

Proof: (1) $\phi(1) = 1 \Rightarrow 1 \in \ker \phi$; if $\phi(a) = 1$ and $\phi(b) = 1 \Rightarrow$
 $\phi(ab) = \phi(a) \cdot \phi(b) = 1 \cdot 1 = 1 \Rightarrow \ker \phi$ is closed wrt.
 $a \in \ker \phi \Rightarrow \phi(a^{-1}) = (\phi(a))^{-1} = 1^{-1} = 1 \Rightarrow a^{-1} \in \ker \phi$ } $\ker \phi \subset G$
is a subgroup

(2) $\phi(1_G) = 1_H \Rightarrow 1_H \in \text{Im} \phi$; if $a = \phi(x)$ and $b = \phi(y)$
 $\Rightarrow ab = \phi(x)\phi(y) = \phi(xy) \Rightarrow ab \in \text{Im} \phi \Rightarrow \text{Im} \phi$ closed wrt.
If $a = \phi(x) \Rightarrow a^{-1} = (\phi(x))^{-1} = \phi(x^{-1}) \in \text{Im} \phi$ } $\text{Im} \phi \subset H$
is a subgroup

Normal subgroup

Definition

Let G be a group. A subgroup $H \subset G$ is a **normal subgroup** if for any $h \in H$, $g \in G$ we have

$$ghg^{-1} \in H.$$

Notation: $H \trianglelefteq G$.

Proposition

Let $\phi : G \rightarrow H$ be a group homomorphism. Then $\ker \phi \trianglelefteq G$ is a normal subgroup in G .

Proof! Let $h \in \ker \phi$, $g \in G$. Need to show: $ghg^{-1} \in \ker \phi$

$$\phi(ghg^{-1}) = \phi(g) \underbrace{\phi(h)}_{h \in \ker \phi \Rightarrow 1} \phi(g^{-1}) = \phi(g) \cdot 1 \cdot \phi(g^{-1}) = \phi(g) \cdot \phi(g^{-1}) = \phi(g \cdot g^{-1}) = \phi(1) = 1$$

ϕ a homomorphism

$$\Rightarrow ghg^{-1} \in \ker \phi \Rightarrow \ker \phi \trianglelefteq G$$
